

PRATIM BAIDYA

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EXPERIENCE

IoTall Labs Private Limited / Kolkata / Data Analyst / Jan – Jun 2025

- Worked on building and optimizing computer vision models for gesture and posture detection. Used libraries like OpenCV and MediaPipe, Keras, Sk-Learn, Matplotlib, Seaborn. Applied callbacks, early stopping and data augmentation techniques to reduce overfitting. Achieved closed to 90% accuracy.
- Developed an audio classification model for detecting and classifying different sounds inside a house using pretrained YAMNet model. Used libraries like Keras, Librosa, Sk-Learn, Matplotlib, Seaborn.
- Conducted product and market analysis for IoT products and reporting.
- Built automated Post and Messaging for different social media platforms using Python Selenium library for digital marketing.
- Explored small vision language models for image generation and video description generation for digital marketing.

EDUCATION

- BS Degree Program | IIT Madras | Data Science | Present CGPA – 8.34 | 2023 – Present [\[Progress\]](#)
- B. Tech | MAKAUT | Mechanical Engineering | CGPA – 9.05 | 2019 – 2022 [\[Certificate\]](#)

PROJECTS

- Ecommerce Data Analysis. [\[Link\]](#)
 - Queries on a large ecommerce dataset of 7 different tables from Python Notebook using MySQL
 - Used Pandas for basic manipulations on the query results Matplotlib and Seaborn for plotting.
- Heart Disease Classification. [\[Link\]](#)
 - Libraries used – Sk-Learn, NumPy, Pandas, Matplotlib, Seaborn
 - Analysis of different feature to look for trends and patterns among them.
 - ML Models – Logistic Regression, K-NN, Random Forest (with tuned hyper-parameters)
 - Model evaluation with Classification report, Confusion Matrix, ROC-AUC Curve.
- Brain Cancer Classification. [\[Link\]](#)
 - Libraries used – PyTorch, Sk-Learn, NumPy, Pandas, Matplotlib.
 - Trained a custom CNN and pre-trained Resnet50 model with 94% accuracy on a balanced dataset.
 - Used callback methods, early stopping and data augmentation techniques to reduce over fitting.
- Sarcasm Detection from News Headlines. [\[Link\]](#)
 - Libraries used – PyTorch, Sk-Learn, Matplotlib
 - Used one of the pretrained BERT model and its tokenizer.

SKILLS

- General – Data cleaning, Feature engineering, Data analysis, Machine Learning, Deep Learning
- Python – OOPs, NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, PyTorch, HuggingFace, FastAPI
- Tools - SQL, Excel, Power BI

AREA OF STUDY

- Probability and Distributions; Descriptive and Inferential Statistics; Linear Algebra.
- Machine Learning- Liner & Logistic Regression (Regularized, Kernal, Polynomial), K-NN, Neural Networks; Tree Based – Random Forest, Boosting; Unsupervised – K-Means, PCA; Model Evaluations.

COURSES

- Advanced Certificate in Machine Learning and Data Science – IIT Madras [\[Credentials\]](#)
- Applied Data Science Lab - WorldQuant University [\[Credentials\]](#)